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Project Seminar 1 Report

Browser Automation with Natural Language Instructions Utilizing Multimodal Methods

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Thesis context and assignment

- Motivation
  - Pivot to automated interactions with web pages using LLM web agents
  - Web agents and benchmarks suffer from multiple problems
  - Robustness is limited by the grounding methods
- Tasks
  - Survey existing grounding methods, datasets and benchmarks
  - Extend existing dataset
  - Propose novel grounding method optimizing for inference speed and low resources
  - Validate and compare novel method on the dataset

The thesis was based mainly on these papers

- MIND2WEB: towards a generalist agent for the web
  - One of the first specialized grounding methods
- GPT-4V(ision) is a generalist web agent, if grounded
  - Grounding methods are a significant bottleneck for web agents
- Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents
  - Grounding method based purely on visual input

What goals have been reached…

- 25+ existing approaches have been surveyed and summarized

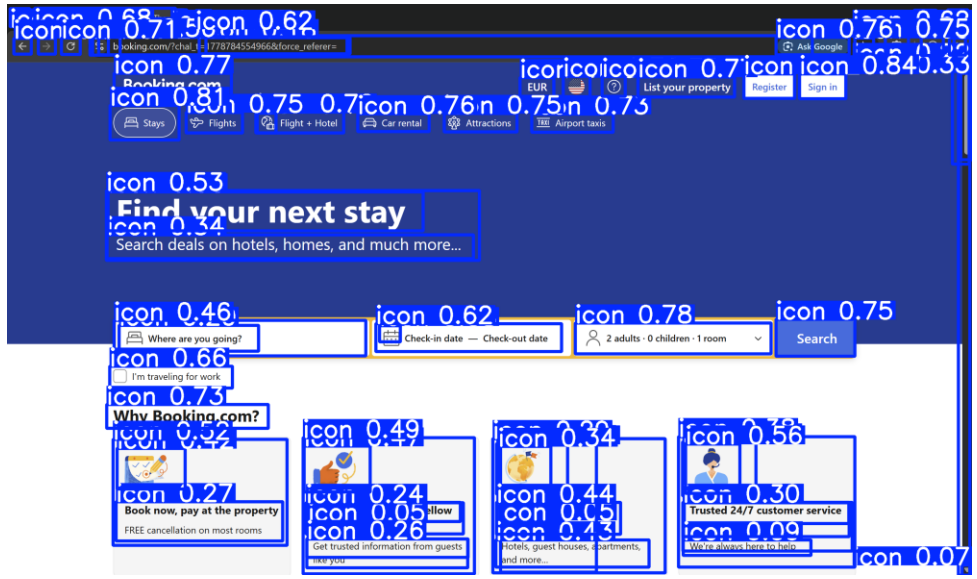
Method	Text	Vision	Approach
MindAct	✗	✓	DeBERTa to choose top-k candidates → multi-choice
$\text{SeeAct}_{\text{Choice}}$	✓	✓	multimodal MindAct
$\text{SeeAct}_{\text{Annotation}}$	✓	✓	Set-of-Mark (mark interactable elements)
OpAgent	✓	✓	finetuned Qwen2.5-VL-7B
UGround	✗	✓	custom LLaVA model
Aria-UI	✗	✓	25B visual mixture-of-expert grounding model
AdaptAgent	✓	✓	few-shot demonstrations or meta-learning
Plan-And-Act	✓	✓	finetuned LLaMA-3.3-70B-Instruct
Auto-GUI	✗	✓	visual encoder
Aguvis-7B	✗	✓	custom visual model
WebWISE	✓	✗	filtered DOM
WILBUR	✓	✗	In context examples
Omniparser	✓	✓	visual model for Set-of-Mark + OCR
WebVoyager	✗	✓	Set-of-Mark using GPT-4V-ACT
Seeclick	✗	✓	custom visual model
WebGUM	✓	✓	encoder-decoder, based on ImageNet-21K
UI-Venus	✗	✓	MoE model trained using reinforcement learning
WALT	✓	✗	Generating and iterating tools

What goals have been reached…

- The thesis introduction
  - Motivation
  - Terminology
  - Previous works
  - Problems of agentic workflows
- The thesis methodology
  - Candidate generation
  - Action prediction
    - Text-first approach
    - Image-first approach

What goals have been reached…

- Set of online web sites for preliminary testing
- Library demos
  - Playwright – access website, download DOM and parse
  - Transformers – download model and analyze screenshot locally
  - Ultralytics – detect UI elements using YOLO model



…what is yet to come…

- May – June 2026 – Proof of concept of the novel method
- July 2026 – Dataset collection
- August – September 2026 – Implementation of the novel method, finetuning
- Winter 2026 - Evaluation of implemented method, comparison with existing methods
- Spring 2027 - Writing and finishing paper

…and what are the limitations

- The volatility of online websites
- The lack of unified benchmark for web agents
- LLM finetuning is resource heavy
- Only a small part of existing models is open source, many are proprietary